NAVRACHANA UNIVERSITY

Teaching Plan

Software Engineering (CS330)

Credit : 3

<table>
<tr><td>Faculty</td><td>
<table>
<tr><th>Name</th><th>Contact</th><th>Office Hours</th><th>Office Location</th></tr>
<tr><td>Ujwala Petigara</td><td>ujwala.petigara@nuv.ac.in</td><td></td><td></td></tr>
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<tr><td>School</td><td>SET</td></tr>
<tr><td>Semester</td><td>Autumn (July to Dec 2025)</td></tr>
<tr><td>Lecture time & Weedays</td><td>
<table>
<tr><td>SECTION_1</td><td>Wednesday</td><td>14:00:00-15:00:00</td><td>07-07-2025 TO 14-11-2025</td></tr>
<tr><td></td><td>Thursday</td><td>12:00:00-13:00:00</td><td>07-07-2025 TO 14-11-2025</td></tr>
<tr><td></td><td>Monday</td><td>12:00:00-13:00:00
12:00:00-13:00:00</td><td>07-07-2025 TO 07-09-2025
08-09-2025 TO 14-11-2025</td></tr>
<tr><td>SECTION_2</td><td>Wednesday</td><td>10:00:00-11:00:00</td><td>07-07-2025 TO 14-11-2025</td></tr>
<tr><td></td><td>Friday</td><td>12:00:00-13:00:00</td><td>07-07-2025 TO 14-11-2025</td></tr>
<tr><td></td><td>Tuesday</td><td>11:00:00-12:00:00
11:00:00-12:00:00</td><td>07-07-2025 TO 07-09-2025
08-09-2025 TO 14-11-2025</td></tr>
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<tr><td>Pre-requisite</td><td></td></tr>
<tr><td>Co-requisite</td><td></td></tr>
<tr><td>Anti-requisite</td><td></td></tr>
<tr><td>Course Description</td><td>This course is aimed at helping students build up an understanding of how to develop a software system from scratch by guiding them thru the development process and giving them the fundamental principles of system development with object oriented technology using UML. The course will initiate students to the different software process models, project management, software requirements engineering process, systems analysis and design as a problem-solving activity, key elements of analysis and design, and the place of the analysis and design phases within the system development life cycle</td></tr>
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<tr><td>Course Objectives</td><td>This course is aimed at helping students build up an understanding of how to develop a software system from scratch by guiding them thru the development process and giving them the fundamental principles of system development with object oriented technology using UML. The course will initiate students to the different software process models, project management, software requirements engineering process, systems analysis and design as a problem-solving activity, key elements of analysis and design, and the place of the analysis and design phases within the system development life cycle</td></tr>
<tr><td>Course Outcomes</td><td>Student will be able to
1. To write and survey solution for multidisciplinary case-study to choose appropriate Process model.
2. To design various system models with respect to different approaches.
3. To implicate various design specification of software.
4. To perform Cost/Effort estimation Planning.
5. To write different test cases according to system design.</td></tr>
<tr><td>Course Typology</td><td>Theory</td></tr>
<tr><td>Course Outline</td><td>As mentioned below in detailed Session Plan below</td></tr>
<tr><td>Pedagogy</td><td>Theoretical Foundations, Software Development Practices, Collaborative Learning, Continuous Assessment and Feedback</td></tr>
<tr><td>Expectations from Students</td><td>Students should showed the interest in the subject and try to understand the concepts thro’ regular revision and attending labs and lectures 100 % and practice programs given in assignments</td></tr>
<tr><td>Assessment / Evaluation</td><td></td></tr>
<tr><td>Project / Assignment Details</td><td>Students should showed the interest in the subject and try to understand the concepts thro’ regular revision and attending labs and lectures 100 % and practice programs given in assignments</td></tr>
<tr><td>Course Material</td><td>Text books:
1. Roger Pressman, “Software Engineering: A Practitioners Approach”, 6thEdition, McGraw Hill, 2010.
2. Ian Somerville, “Software Engineering”,9 th Edition, Addison Wesley, 2011
3. James F. Peters and WitoldPedrycz, “Software Engineering – An Engineering Approach”, Wiley, 2000.
Reference books:
1. Roger S.Pressman, Software engineering- A practitioner’s Approach, McGraw-Hill International Editions
2. Ian Sommerville, Software engineering, Pearson education Asia
3. Pankaj Jalote, Software Engineering – A Precise Approach Wiley
4. Software Engineering Fundamentals by Ali Behhforoz & Frederick Hudson OXFOR</td></tr>
<tr><td>Additional Information</td><td>The course will be useful to understand and implement work on project</td></tr>
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Session Plan	Topic Title	Session No.	Topic & Subtopic Details	Readings, Cases, etc.	Activities	Important Dates
		0				
	INTRODUCTION	1	Introduction to Software Engineering Software Components, 8 Software Characteristics, Software Engineering Processes, Similarity and Differences from Conventional Engineering Processes			
		2	Software Quality Attributes. Software Development Life Cycle (SDLC) ModelsWater Fall Model, Prototype Model Spiral Model			
		3	Evolutionary Development Models,Iterative Enhancement Models			
		4	Typical Application for each model, Agile methodology, Process and Project Metrics			
	SYSTEM ANALYSIS	5	System Requirement			
		6	System Requirement			
		7	Stakeholders			
		8	Techniques for information gathering			
		9	Techniques for information gathering			
		10	Validating the requirements			
		11	Validating the requirements			
		12	Feasibility Analysis			
		13	Tests for feasibility			
		14	Cost Benefit Analysis			
		15	Feasibility report			
		16	The system Proposal Data flow diagram and UML analysis diagrams			
		17	The system Proposal Data flow diagram and UML analysis diagrams			
	SOFTWARE DESIGN	18	Software Design Specification– Abstraction Modularity			
		19	Effective modular design			
		20	Cohesion and Coupling			
		21	Example of code for cohesion and coupling			
		22	User Interface Design – Human Factors,			
		23	Interface standards– User Interface Design Process Application Architecture for Information System			
		24	Interface standards– User Interface Design Process Application Architecture for Information System			
		25	Deployment using UML diagrams			
		26	Component and deployment diagram for various architectures Design Patterns Gang of four patterns			
	SOFTWARE QUALITY ASSURANCE	27	Software estimation			
		28	Empirical estimation models			
		29	Cost/Effort estimation Planning			
		30	Work breakdown Structure, Gantt Chart			
		31	Evaluate schedule and cost variance Risk Management Risk Identification			
		32	Risk Assessment			
		33	Risk Projection			
		34	RMMM			
		35	Software Configuration Management			
		36	SCM process, version and change control Quality metrics			
	SOFTWARE TESTING	37	Software Testing Basic concept and terminology			
		38	Verification & validation			
		39	White Box Testing-, Black Box Testing			
		40	Integration, Validation and system testing			
		41	Object Oriented testing methodsClass Testing, Interclass testing			
		42	Object Oriented testing methodsClass Testing, Interclass testing			
		43	Interclass testing,			
		44	testing architecture,			
		45	Behavioural testing			
Publish Status	Published					